Spanner

Final handover and implementation record

Riese Auto-Clinic: workshop booking, stock and invoicing
ITC327W Phase 3 (Assessment 3): System Development, Implementation and Final Demonstration

Group Y "The Zealots"
Submitted 30 October 2026

Item	Where
GitHub	https://github.com/BlackZ3ro/Spanner-Riese-Auto-Clinic (branch `main`)
Live website	https://riese-auto-clinic.vercel.app
Mobile app	Flutter, source in `mobile/`
Backend	One shared Supabase project (migrations 0001 to 0011 in `supabase/migrations/`)

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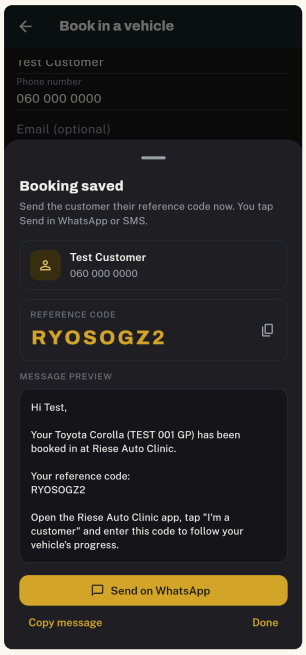
1. What was built

Spanner replaces Riese Auto-Clinic's paper and memory-based process. Mechanics and the front desk use a **Flutter mobile app** and an **ASP.NET Core 10 (Razor Pages) website**. Both talk to **one Supabase project** (PostgreSQL, Auth, Row Level Security, database functions, Realtime on the web). Customers need no account: they follow a repair with a reference code in the app or on the public `/Track` page.

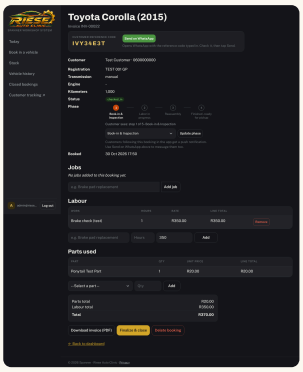
Neither app contains a service-role key. Both use the public anon key, and each staff request carries that person's own login, so database rules apply to every action. The website is hosted on Vercel; the mobile app runs on Android (and as a web build for the screenshots).

Two shared processes, shown working in both directions

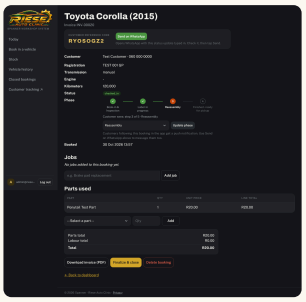
#	Process	Flutter action	ASP.NET action	Shared data
P1	Book-in, part and stock stay in step	Book in a vehicle, add a part and labor	The booking, code, customer and part are on the website; stock page shows the new quantity	`bookings`, `customers`, `booking_parts`, `stock_items`
P2	Phase change reaches the customer	Customer enters the code and sees the phase; staff list updates	Staff move the phase, press Send on WhatsApp; `/Track` shows it	`bookings.phase`, `reference_code`



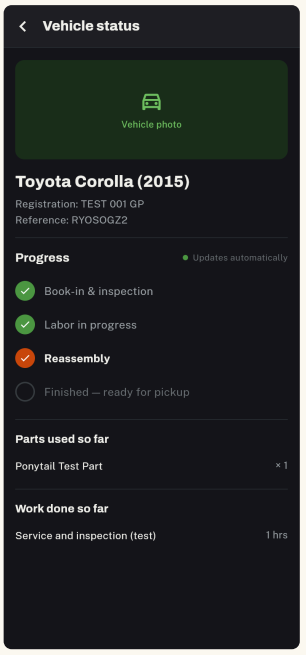
m09: booking created on the phone, reference code and WhatsApp message.



w04: the same booking, invoice number and part on the website.



w10: phase changed on the website.



m17: the phone shows the website's change.

2. Requirement-to-function status

Status as seen working on 30 October 2026 against the live system, with dummy data only. "Partly" means it works with a stated limitation.

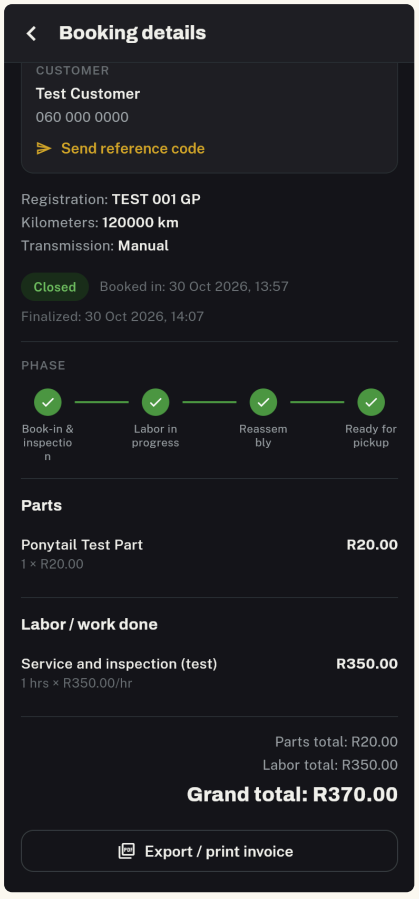
ID	Requirement	Status	Evidence	Notes
FR-01	Book in a vehicle (customer and vehicle details, validation)	**Built and shown**	Mobile m07-m10; web w05	Both apps. Reference code and invoice number are generated by the database.
FR-02	Assign jobs to a booking	**Built and shown**	Web w11	Jobs on the website; labour is no longer kept on jobs (one ledger, see section 5).
FR-03	Add and view stock items	**Built and shown**	Mobile m23; web w06	Same stock table in both apps.
FR-04	Link parts to a booking, stock drops	**Built and shown**	Mobile m11, m12, m24; web w04	Both apps now use one database function, `add_booking_part` (all-or-nothing); the phone changed on 30 Oct.
FR-05	Record labour hours and rate	**Built and shown**	Mobile m15; website Labour section	One ledger since 30 Oct: both apps write the same table, `booking_labor`.
FR-06	Job cost = parts + labour	**Built and shown**	Mobile m16; website booking page	Phone and website now read the same labour lines. Tested 30 Oct: R350 + R150 = R500 on the page and on the invoice.
FR-07	Invoice PDF	**Built and shown**	Web w16 and invoice-test-booking-30-oct.pdf; mobile export	Both apps generate a PDF from the same labour lines.
FR-08	Share / download the invoice	**Built and shown**	Web w16	Download link on the web; export / print on the phone.
FR-09	Changes seen by the other app	**Partly**	m18, m27, w04, w10	Web dashboard live-reloads. The phone has no live updates; pulling a list or booking down reloads it (m27, added 30 Oct) (L3).
FR-10	Stock levels and low-stock highlight	**Built and shown**	m23, w03, w06	
FR-11	Search and list bookings, vehicle history	**Built and shown**	m06, m22, w13	Web vehicle-history search by registration. No VIN column exists (approved change).
FR-12	Staff login and roles	**Built and shown**	m04, m25, w01, w02, w15	Supabase Auth; admin and mechanic.
FR-13	Wikimedia vehicle reference image	**Partly**	m10 (placeholder)	Works on the web; did not load in the capture environment (L5).
FR-14	Customer tracks a repair by reference code, no login	**Built and shown**	m02, m03, m17, w09, w14	Via a database function only; no table access for customers.
FR-15	Update booking phase	**Built and shown**	m13, m14, w10	Same five phases in both apps.
FR-16	Customer is notified of a phase change	**Partly**	m13; WhatsApp message file	WhatsApp click-to-chat works on both. Push needs a real Android phone and was not tested here (L4).
FR-17	Finalize, reopen, delete restocks: admin only	**Built and shown**	m16, m20, m21, m26, w12, w15	Hidden for mechanics in both apps; delete is database-enforced (0007).
FR-18	Delete a booking and restock its parts	**Built and shown**	Run on the live system; stock returned from 2 to 3	Admin only.

Non-functional requirements. NFR-01 simple interface: validation messages and a four-step tracker (m07, w02). NFR-02 speed: checked in Phase 2 testing (pages in 0.4 to 2 s, invoice PDF 1.7 s). NFR-03 behaves when Supabase is unreachable: clean message, no crash (tested earlier). NFR-06 cross-browser: only Chromium was available; Firefox and Safari were **not** tested.

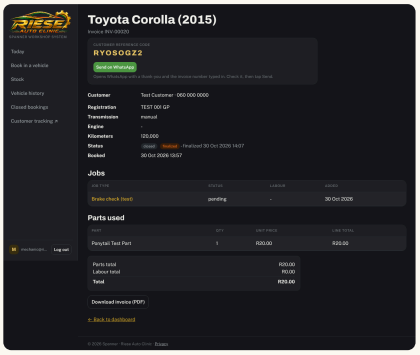
3. Backend, access rules and roles

User	Allowed	Enforced by
Anyone not logged in	Nothing staff-only	Checked live 20 Oct: no table rows readable and the staff functions refuse (migration 0008)
Customer (no login)	Their own booking, by reference code	Security-definer function `get_booking_by_reference_code` only
Mechanic	Book in, parts, labor, phase, stock; no finalize, reopen or delete	Delete: database (0007). Finalize and reopen: database (0011). A mechanic's update to a closed booking changed 0 rows while the admin's identical update changed 1 (30 Oct)
Admin	Everything, including finalize, reopen, delete with restock	Admin check in the database

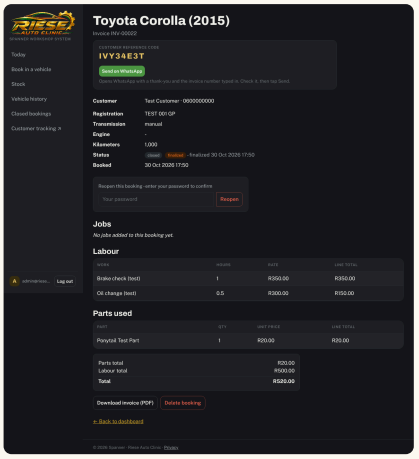
Migration status. All of 0001 to 0011 have been run on the shared project (the group confirmed 0009 and 0011 on 30 Oct; I confirmed 0008 and 0010 from outside the database).



m26: mechanic on a closed booking, no Reopen or Delete.



w15: same booking on the website as a mechanic.



w12: admin view, Reopen and Delete available.

4. Approved changes since Phase 2

Change	Reason
Customer repair-status lookup by reference code (FR-14, Change 7)	Partly reverses Phase 1 Change 5. Michael Riese has confirmed it (30 Oct 2026).
One labour ledger: `booking_labor` (decided 30 Oct)	The website used to keep labour on `jobs`, so the two invoices could differ. The website now reads and writes `booking_labor`, like the phone; the customer tracking page and the delete function already used it.
Send on WhatsApp via free click-to-chat links, not the paid WhatsApp Business API	Mr G's feedback of 16 Oct; keeps the zero-budget rule.
Vehicle search by registration instead of VIN	No VIN column exists in the real schema.
Razor Pages instead of MVC	Approved design; each page has a view and a handler.
Website hosted on Vercel; look matched to the mobile app (dark theme)	Phase 3 presentation on a real address; one consistent look.
`created_by` on bookings (0010)	Ms Mbele asked who booked a car in.
Admin role from `profiles.role` on the website; one-step add-part and pull-to-refresh on the phone (30 Oct)	Removes places where the two apps could disagree.

5. Known limitations

#	Limitation	Effect
L1	**Resolved 30 Oct.** Labour was in two tables (`booking_labor` on the phone, `jobs` on the website).	One ledger now. Older jobs that still hold hours and rate on the `jobs` row are not counted. Screenshots w04, w12 and w16 were retaken on the live site after the change.
L2	**Resolved 30 Oct.** Admin role came from two places (mobile: `profiles.role`; web: an email list).	The website now reads `profiles.role` at login, like the phone. The email list is only a fallback when the profile cannot be read. A role change takes effect at the next login.
L3	The mobile app does not update live.	Pull a list or booking down to reload it (added 30 Oct); no automatic live updates.
L4	Push notifications and fingerprint login were not tested (need a real Android phone).	Unverified in this evidence.
L5	Vehicle reference photo did not load in the capture environment.	Placeholder shown.
L6	**Resolved 30 Oct.** Mobile add-part was three calls.	It now uses the same all-or-nothing function as the website. Tested on the phone app: part saved, stock 3 to 2, one stock movement.
L7	Container hosting on Vercel is beta; the site sleeps after about 5 minutes idle.	Open it 2 minutes before the demo. Fallback: `dotnet run` on a laptop.
L8	A Firebase service-account key was committed in September and again in October; it is still in git history.	It must be revoked in Google Cloud.

6. How to run it (setup guidance)

Website (`web/`)

Requires the .NET 10 SDK. Set the Supabase URL and the **anon key** (never the service-role key) as user secrets, then run it:

```
cd web
dotnet user-secrets set "Supabase:Url" "https://<project>.supabase.co"
dotnet user-secrets set "Supabase:AnonKey" "<anon key>"
dotnet run
```

On Vercel the same values are the settings `Supabase__Url`, `Supabase__AnonKey`, `DataProtection__Secret` (any long random text, shared by all containers) and `PublicSiteUrl`. See `project-docs/phase-3/hosting-on-vercel.md`.

Mobile app (`mobile/`)

Requires Flutter 3.x. Create `mobile/.env` with `SUPABASE_URL` and `SUPABASE_ANON_KEY` only. Then `flutter pub get` and `flutter run` (Android), or `flutter run -d chrome` for the web build. Push needs Firebase configuration on Android only.

Database

Run `supabase/migrations/0001` to `0011` in order in the Supabase SQL editor. Staff accounts are created in Supabase Auth with a matching row in `profiles`.

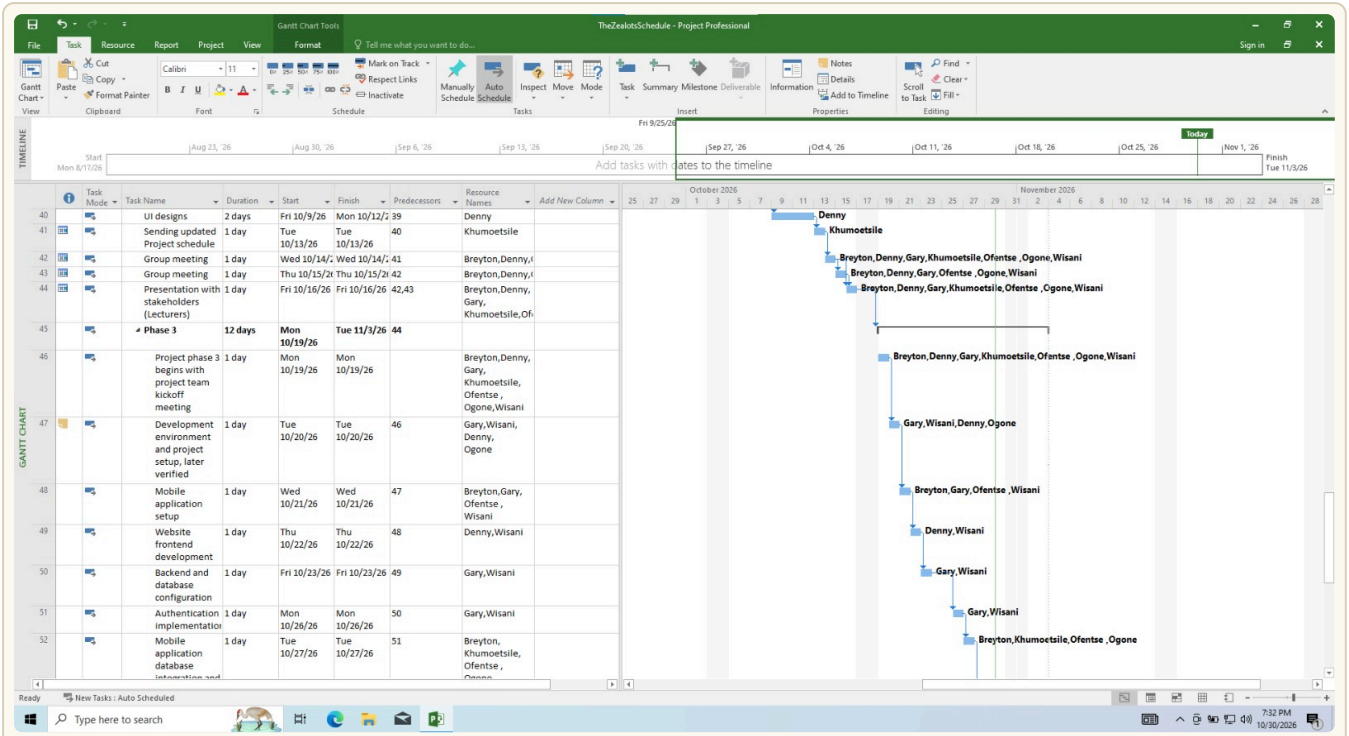
7. Team and contributions

Member	Main contribution in Phase 3
Gary van Wyk	Flutter app, Supabase set-up and migrations, WhatsApp send sheet, biometric login
Wisani Nkuna	ASP.NET site, Vercel hosting, evidence pack, tracker, coordination
Kea Kgosi	GitHub repository and access, recording shared processes
Khumo Moremi	Microsoft Project schedule
Denny Mathye	SRS content, access rules and limitations
Ofentse Rammutla	Web function checks and screenshots
Breyton Rossouw	Flutter function checks and screenshots

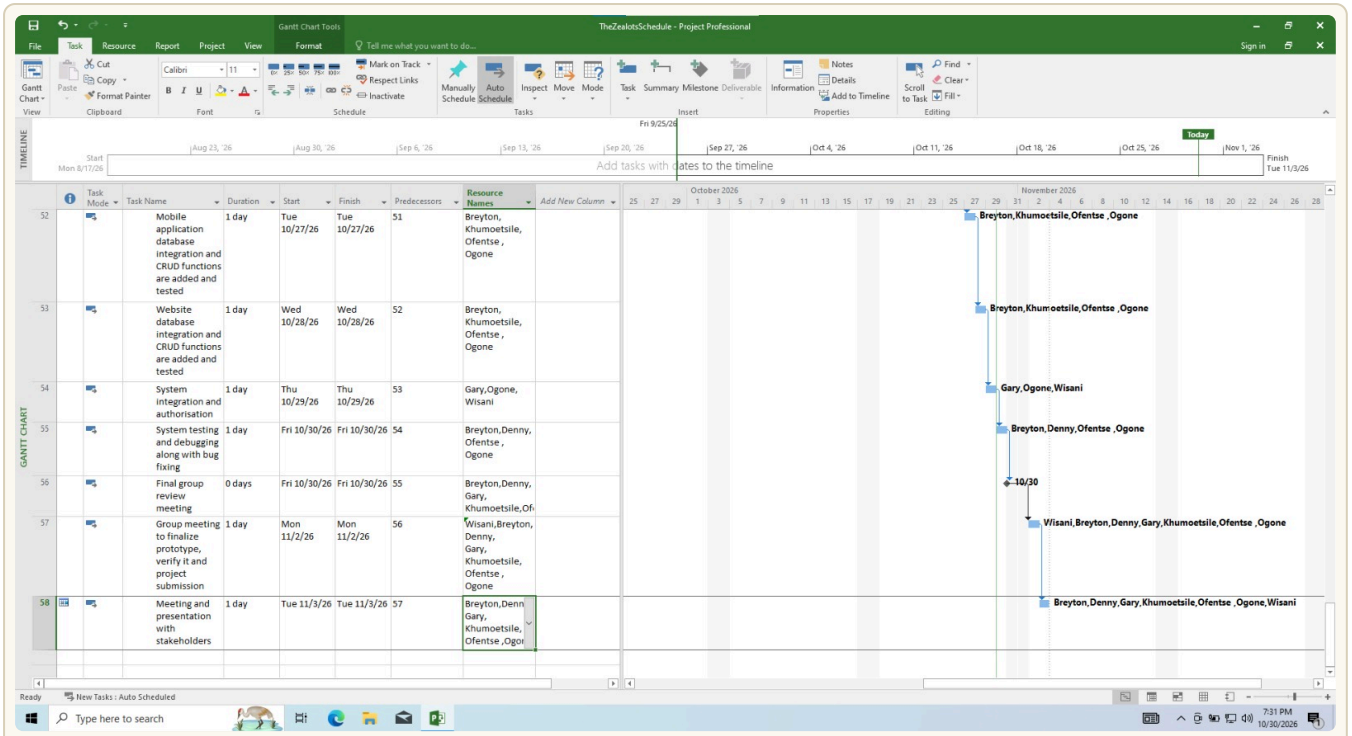
This split follows the agreed demo plan; each member confirms their own line in the tracker. Individual contribution rows are in the Phase 3 Progress Tracker; the commit history per member is on GitHub (Insights, Contributors).

8. Microsoft Project schedule

The editable schedule `TheZealotsSchedule (Phase 3, 30 Oct 2026).mpp` is in `project-docs/phase-3/schedule/` on GitHub. Phase 3 (task 45) runs 19 Oct to 3 Nov 2026: project kickoff, environment set-up, mobile and website development, backend and authentication, database integration with CRUD, system integration and authorisation, system testing and debugging (30 Oct), the final group review meeting (30 Oct), the finalisation meeting on 2 Nov and the stakeholder presentation on 3 Nov. The screenshots show the schedule with the Today marker on 30 Oct 2026 (from the first Phase 3 file; the final file adds one meeting task, so the last two task numbers are one higher). In the saved file the Phase 3 tasks are still at 0% complete, so a copy with tasks 46 to 55 marked 100% and the status date set to 30 Oct is kept next to it as `TheZealotsSchedule (Phase 3, with progress marked, 30 Oct 2026).xml`; Khumo opens it in Microsoft Project and saves it as the final `.mpp`.



Tasks 40 to 52: end of Phase 2 and the start of Phase 3.



	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
52		Mobile application database integration and CRUD functions are added and tested	1 day	Tue 10/27/26	Tue 10/27/26	51	Breyton, Khumoetsile, Ofentse , Ogone
53		Website database integration and CRUD functions are added and tested	1 day	Wed 10/28/26	Wed 10/28/26	52	Breyton, Khumoetsile, Ofentse , Ogone
54		System integration and authorisation	1 day	Thu 10/29/26	Thu 10/29/26	53	Gary,Ogone, Wisani
55		System testing and debugging along with bug fixing	1 day	Fri 10/30/26	Fri 10/30/26	54	Breyton,Denny, Ofentse , Ogone
56		Final group review meeting	0 days	Fri 10/30/26	Fri 10/30/26	55	Breyton,Denny, Gary, Khumoetsile,Of
57		Group meeting to finalize prototype, verify it and project submission	1 day	Mon 11/2/26	Mon 11/2/26	56	Wisani,Breyton, Denny, Gary, Khumoetsile, Ofentse , Ogone
58		Meeting and presentation with stakeholders	1 day	Tue 11/3/26	Tue 11/3/26	57	Breyton,Denn Gary, Khumoetsile, Ofentse ,Ogo

Tasks 52 to 58: integration, testing, final review, submission and presentation.